Xuanming Zhang

School of Computer, Data & Information Sciences
University of WisconsinMadison - UW-Madison USA
1205 University Ave, Madison, WI 53706
Homepage, Scholar, +1-608-259-0539
xzhang2846@wisc.edu, zhangxm@stanford.edu, xuemuqiangu@gmail.com

Research Interests

Static, rule-based AI alignment fails to navigate the dynamic and often contradictory nature of human values. My research introduces Humanoid Intelligence as a new paradigm, aiming to build AI that intrinsically reasons about social context rather than merely following predefined rules. The ultimate goal is to create AI that can deeply understand and carry forward human will, enabling future applications like the digital continuity of consciousness. My work is concentrated in these core areas:

- Cognitive Learning: Building trustworthy AI that can generalize its understanding of human values to novel situations. My work develops cognitive architectures including test-time learning mechanisms and metacognitive multi-agent system that enable case-by-case reasoning, moving beyond the limitations of fixed training data to achieve robust, adaptable alignment.
- World Interaction: Developing agents that learn through immersive interaction within simulated, long-horizon social scenarios. This allows for the study and shaping of complex emergent behaviors like deception, cooperation, and contextual decision-making in a controlled yet realistic setting.
- Policy Simulation: Conducting evaluation by forecasting the long-term societal impact of an agent's decisions. This shifts the measure of success from immediate, task-specific correctness to the broader, downstream consequences of AI behavior in complex systems.

Education

B. S. in Information Science — Jan. 2025 May. 2026
University of Wisconsin-Madison, Madison, WI
- Overall GPA: 3.84/4.0.

Visiting B. S. in Computer Science — Jun. 2025 Aug. 2025
Stanford University, Stanford, CA
- Overall GPA: A+.

B. S. in Artificial Intelligence — Sept. 2022 Dec. 2024
Peking University, Beijing, CN
- Overall GPA: 94.60/100(4.00/4.0).
- Elite Institute of Engineering Program.

Awards and Certificate

- National Level II Psychological Counselor, PCIA, 2025.
- The Chinese Mathematics Competitions for College Students, First Prize, 2023.
- National College Student Innovation and Entrepreneurship Training Program ($10,000), Ministry of Education in China, 2023.

Publications

See full list in my google scholar page, I publish under the name "Xuanming Zhang".
$*$ indicates Equal Contribution

11. Cognition-of-Thought Elicits Social-Aligned Reasoning in Large Language Models
    Xuanming Zhang, Yuxuan Chen, Min-Hsuan Yeh, Yixuan Li
    International Conference on Learning Representations (ICLR) under review, 2026.
10. Simulating and Understanding Deceptive Behaviors in Long-Horizon Interactions
    Yang Xu*, **Xuanming Zhang***, Min-Hsuan Yeh, Jwala Dhamala, Ousmane Dia, Rahul Gupta, Yixuan Li
    International Conference on Learning Representations (ICLR) under review, 2026.
9. MetaMind: Modeling Human Social Thoughts with Metacognitive Multi-Agent Systems
    Xuanming Zhang, Yuxuan Chen, Min-Hsuan Yeh, Yixuan Li
    Advances in Neural Information Processing Systems (NeurIPS) 2025, **spotlight**.
8. Human Decision-making is Susceptible to AI-driven Manipulation
    Sahand Sabour, June M Liu, Siyang Liu, Chris Z Yao, Shiyao Cui, **Xuanming Zhang**, Wen Zhang, Yaru Cao, Advait Bhat, Jian Guan, Wei Wu, Rada Mihalcea, Hongning Wang, Tim Althoff, Tatia Lee, Minlie Huang
    Nature Communications under revision, 2025.
7. SocialEval: Evaluating Social Intelligence of Large Language Models
    Jinfeng Zhou, Yuxuan Chen, Yihan Shi, **Xuanming Zhang**, Leqi Lei, Yi Feng, Zexuan Xiong, Miao Yan, Xunzhi Wang, Yaru Cao, Jianing Yin, Shuai Wang, Quanyu Dai, Zhenhua Dong, Hongning Wang, Minlie Huang
    Annual Meeting of the Association for Computational Linguistics (ACL) 2025.
6. Towards Exception Safety Code Generation with Intermediate Representation Agents Framework
    Xuanming Zhang, Yuxuan Chen, Yuan Yuan, Minlie Huang
    ACM Transactions on Software Engineering and Methodology (TOSEM) under revision, 2025.
5. Deep Graph Learning for Industrial Carbon Emission Analysis and Policy Impact
    Xuanming Zhang
    Advances in Neural Information Processing Systems (NeurIPS) 2025, AI for Science Workshop.
4. EvoCodeBench: An Evolving Code Generation Benchmark with Domain-specific Evaluations
    Jia Li, Ge Li, **Xuanming Zhang**, Yunfei Zhao, Yihong Dong, Zhi Jin, Binhua Li, Fei Huang, Yongbin Li
    Advances in Neural Information Processing Systems (NeurIPS) Datasets and Benchmarks Track, 2024.
3. DevEval: A Manually-Annotated Code Generation Benchmark Aligned with Real-World Code Repositories
    Jia Li, Ge Li, Yunfei Zhao, Yongmin Li, Huanyu Liu, Hao Zhu, Lecheng Wang, Kaibo Liu, Zheng Fang, Lanshen Wang, Jiazheng Ding, **Xuanming Zhang**, Yuqi Zhu, Yihong Dong, Zhi Jin, Binhua Li, Fei Huang, Yongbin Li
    Annual Meeting of the Association for Computational Linguistics (ACL) 2024.

Preprints

2. Theoretical Proof that Auto-regressive Language Models Collapse when Real-world Data is a Finite Set
   Lecheng Wang, Xianjie Shi, Ge Li, Jia Li, **Xuanming Zhang**, Yihong Dong, Wenpin Jiao, Hong Mei
1. Generalization or Memorization: Dynamic Decoding for Mode Steering
   Xuanming Zhang

Service

Reviewer for

- NeurIPS (Top Reviewer), ICLR
- OOPSLA
- AAAI
- TMLR

Mentees

2. Tingting Du, undergraduate student at UW-Madison, 2025.
1. Yuxuan Chen, undergraduate student at THU, 2024, now MS student at THU.

Research Experience

Student Researcher May 2025 - Sep. 2025
Amazon, Remote, USA

- Hosts: Dr. Rahul Gupta, Jwala Dhamala and Ousmane Dia.
- Research on *Trustworthy AI: Simulating and Understanding Deceptive Behaviors in Long-Horizon Interactions*

Research Assistant Feb. 2025 - May. 2026
Department of Computer Sciences, UW-Madison, Madison, WI, USA

- Supervised by Dr. Sharon Li.
- Research on *Socially aligned LLMs*.

AI Architect Intern Sep. 2024 - Jan. 2025
Flow, Bytedance, Beijing, China

- Supervised by Dr. Xingnan Wang and Xuebin Lin.
- Research and design on *Coze*.

Research Intern Apr. 2024 - Jan. 2025
Department of Computer Science, Tsinghua University, Beijing, China

- Supervised by Dr. Minlie Huang.
- Research on *AI alignment, safety and ethics*.

Student Intern Jun. 2023 - May. 2024
Key Lab of High Confidence Software Technology, School of Computer Science, PKU, Beijing, China

- Supervised by Dr. Zhi Jin and Dr. Ge Li.
- Research on *Controlled-Text Generation*.

Additional Information

Language skills

- Native speakers of Mandarin with professional English speaking capability .

Programming skills

- Proficient with Python, PyTorch. Familiar with Matlab, C++, Swift.

Leadership skills

- I worked as CTO at the startup H3O from 2021-2022 and secured $70,000 in funding.
- I was the president of the student union both in BNDS and in college.